

Business Problem Investigation

Problem 1 — Why does Furniture have high sales but low profit?

Investigation: The Furniture: Sales vs. Profit by Sub-Category chart shows that all four sub-categories (Chairs, Tables, Bookcases, Furnishings) generate meaningful sales, but their profit bars are minimal or negative in comparison. Chairs has the highest sales among the four, yet its profit is only marginal. Tables is the most severe case, generating a negative profit margin (-9%), meaning the company loses money on every sale in that sub-category once costs are accounted for.

Root cause (from Discount vs. Profit chart): The Average Discount by Category chart shows Furniture carries a noticeably higher average discount than other categories. Combined with the Discount vs. Profit scatter — where Tables and Bookcases cluster at higher discount levels with low or negative profit — this suggests **discounting is a major driver of Furniture's weak profitability**, not just higher product costs.

Conclusion: Furniture isn't a "bad" category in terms of demand — it sells well — but aggressive discounting is eroding margins, especially in Tables and Bookcases.

Problem 2 — Which months drive the strongest and weakest sales?

Investigation: The Monthly Sales Trend chart shows a clear seasonal pattern. Sales start relatively low in January and February, dip further around April–June, then climb sharply from September through December, with November and December representing the strongest months of the year.

Conclusion: The business shows a clear **end-of-year sales surge**, likely tied to holiday shopping season, while Q1–Q2 (particularly spring) is comparatively weak. This has direct implications for inventory planning, marketing spend timing, and staffing — resources should be concentrated toward Q4, with targeted promotions considered for the slower spring months to smooth out performance.

Problem 3 — How are discounts affecting profitability?

Investigation: The Discount vs. Profit by Sub-Category scatter chart shows a general pattern: sub-categories with **higher discount sums cluster toward lower or negative profit**, while sub-categories with minimal discounting show healthier profit. This is reinforced by the Average Discount by Category chart, where Furniture — the category with the worst profit margin — also carries the highest average discount among all categories.

Conclusion: There is a visible relationship between discount level and reduced profitability. While discounting can help drive sales volume, the evidence suggests that **beyond a certain point**,

discounts are cutting directly into profit rather than generating enough extra volume to compensate — particularly in Furniture.